

Intelligent Holographic Antiglare Windshield System with Head-up Display

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One main challenge faced in modern automotive safety design is the glare caused by strong external light. When glare directly hits the driver's eyes, the extreme brightness distribution in the driver's field of view reduces environmental visibility, causing a decrease in driver judgment and reaction speed, which increases the probability of accidents occurring. To reduce traffic accidents caused by glare in a changing environment, an intelligent holographic antiglare windshield system (IHAWS) based on a holographic optical element (HOE) is proposed. The antiglare HOE (aHOE) in IHAWS attenuates the external strong light directly in the driver's field of view to an acceptable level through diffraction and absorption of incident light, thereby alleviating abnormal brightness distribution caused by glare. The HOE in head-up display (hHOE) projects driving information such as speed and navigation to the front of the vehicle, thus reducing safety hazards by minimizing driver distraction. The compatibility between these two types of HOEs requires an intelligent scheduling scheme based on binocular camera detection. The IHAWS proposed in this article attenuates the intensity of light directly shining on the driver's eyes, which inspires new solutions to reduce the incidence of accidents caused by glare.

or image recognition systems.^[13] Due to the fact that these methods extract effective information from videos containing glare through image processing, their main application scenarios are limited in the field of machine vision, mainly serving as an auxiliary or substitute for human drivers. Nevertheless, the interference of glare on the driver's visual perception is still inevitable.

The hardware-based antiglare method mainly targets traditional driving scenarios at present, and alleviates the discomfort caused by glare to human drivers through upgrading and innovating the in-car devices. Typical solutions include glare-free intelligent high beams,^[14,15] night vision systems,^[16] antiglare materials,^[17] and antiglare glasses.^[18] The intelligent high beam eliminates other vehicles from the beam through pixelated adjustment. Obviously, the above scheme can avoid the interference of high beams on other vehicles, but it cannot protect our own drivers from

1. Introduction

The safety design of automobiles attracts increasing attention. For antiglare design in particular, glare slows down the driver's reaction speed and dulls their judgment, which aligns with existing evaluations,^[1] surveys,^[2] and empirical studies.^[3–9]

The current methods used to eliminate glare can be divided into two categories: software-based and hardware-based methods. Among them, software-based glare reduction methods mainly apply in autonomous driving,^[10,11] assisted driving systems,^[12]

the influence of other light sources. The night vision system uses infrared technology, but it is costly and does not conform to human observation habits. Some liquid crystal materials can prevent glare by filtering yellow light, but their environmental adaptability is difficult to meet durability requirements. Antiglare glasses mainly reduce glare through polarization selection and other methods, but they will affect color perception and are not suitable for use at night.

Holographic optical elements (HOE) are designed flexibly, easy to integrate, and suitable for large-scale production, making them ideal for integration into existing glass systems to achieve controllable utilization of light energy. HOE has achieved some successful application cases in the fields of buildings,^[19,20] energy,^[21–25] and automotive head-up display (HUD).^[26–30] But to our knowledge, there has not yet been an application of HOE in the field of automotive antiglare design.

Overall, there are some systems that implement antiglare functions in automobiles. However, providing basic optical solutions for antiglare without affecting the normal operation of HUD remains an urgent issue in the field of automotive safety design.

In this study, we propose an intelligent holographic antiglare windshield system (IHAWS) based on HOE. By integrating the multiplexed HOE designed in different regions into the original windshield, the balance of antiglare effects in a large angle range and a large spectral range is achieved. The intelligent scheduling scheme based on binocular cameras avoids the overlap of the

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working areas of antiglare HOE (aHOE) and the HOE used for head-up display (hHOE) through real-time feedback, ensuring the normal operation of HUD while preventing glare. It is believed that IHAWS will drive the development of safety design in intelligent cockpits.

2. Results and Discussion

2.1. System Architecture

Figure 1 shows the basic architecture and working principle of IHAWS. The driver is located inside the cab and observes the external environment through the windshield. HUD provides necessary driving assistance information to the driver through augmented reality (AR) display in front of the automobile. When strong external light is incident at a specific angle, the light falling near the eye box can cause abnormal brightness distribution in the driver's field of view, resulting in glare that seriously interferes with the driver's visual perception ability. This article aims to develop an antiglare system based on HOE to reduce strong light entering the driver's eye box and avoid interference between the system and AR display through an intelligent scheduling scheme.

The main factors that cause glare during the day are direct sunlight and reflections from the glass exterior walls of buildings. The main factors that cause glare at night are high beams, public lighting, and searchlights. On the one hand, different light sources have different spectra. Figure 1a and b, respectively, show the effects of the intelligent antiglare system on the sunlight during the day and the LED high beam at night. On the

other hand, the position of the light source and the direction of the vehicle's movement are constantly dynamically adjusted, and the angle of external light incidence changes at any time. Therefore, IHAWS needs to have antiglare function for the entire incident angle range and the entire spectral range.

The highlights and potential value of the proposed system are briefly analyzed here. The main function of IHAWS is to suppress glare and reduce driver judgment errors caused by glare, which will improve the all-weather driving safety of cars in complex environments. In addition, its intelligent scheduling capability enables IHAWS to seamlessly integrate with intelligent cockpits containing HUD, which is beneficial for the safety integration design of automobiles. Furthermore, IHAWS has high flexibility in design and scalability, allowing it to be easily transformed into a device that is resistant to high temperatures, UV rays, and collects light energy, thereby enhancing driving comfort, health, and energy sustainability.

2.2. Design of AHOE

Figure 2a is a simulation model of the aHOE, where the orange rectangle represents the windshield, the blue rectangle represents the range of movement of the driver's eyes, the yellow and blue arrows represent the light from the sun and LED, respectively, and aHOE-1 and aHOE-2 are two typical examples, representing the effective parts for attenuating incident light from different directions. The black color represents the coordinate system used in the simulation, and the red arrow indicates the range of external light that can affect the driver.

To achieve antiglare design for large angles and a wide spectrum, the total attenuation coefficient A_c of HOE over the entire spectral range is used as the evaluation index

$$A_c = \sum_{i=1}^{N_s} \left(\sum_{j=1}^{N_{L,i}} \eta_{i,j} \right) / \sum_{i=1}^{N_s} N_{L,i} \quad (1)$$

among them, N_s is the number of samples of wavelengths within the spectral range, $N_{L,i}$ is the number of incident light rays entering the eye box range at a certain wavelength, and $\eta_{i,j}$ is the diffraction efficiency of a certain light ray after HOE modulation. A_c represents the proportion of diffracted light energy to incident light energy, and the higher its value, the better the attenuation effect of HOE on the spectral range of interest at the current incident light angle.

Figure 2b illustrates the design process using a simulation diagram of a certain incident angle and incident wavelength as an example. The XOY plane is the plane where the windshield is located. The blue rectangle in this plane represents the effective aHOE corresponding to the current incident light direction, the yellow rectangle represents the eye box range, the pink and dark blue lines represent the recording light, the cyan line represents the incident external light, and the green line represents the diffracted light. When aHOE is reflective, the light within the angular bandwidth and wavelength bandwidth of the incident light is diffracted back to the outside world, and this part of the light no longer enters the driver's eyes, thereby attenuating the incident light intensity. The red dot represents the landing point of the incident light on the eye box plane when aHOE does not exist.

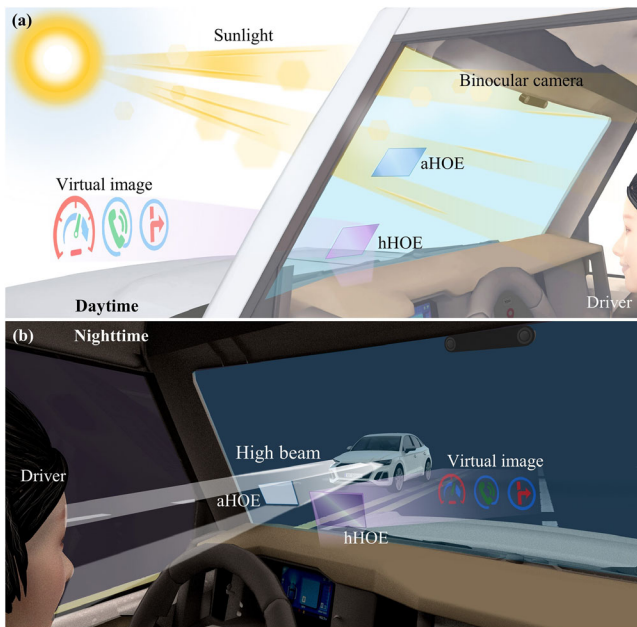


Figure 1. Illustration of the architecture and working principle of proposed system. a) The sunlight that would have entered the driver's eye box is attenuated by aHOE to prevent glare. Since the working areas of aHOE and hHOE do not overlap at this time, all areas of hHOE are working, projecting a virtual image of the full field of view in front of the vehicle. b) The light from the LED high beam that would have entered the human eye is attenuated by aHOE to suppress glare.

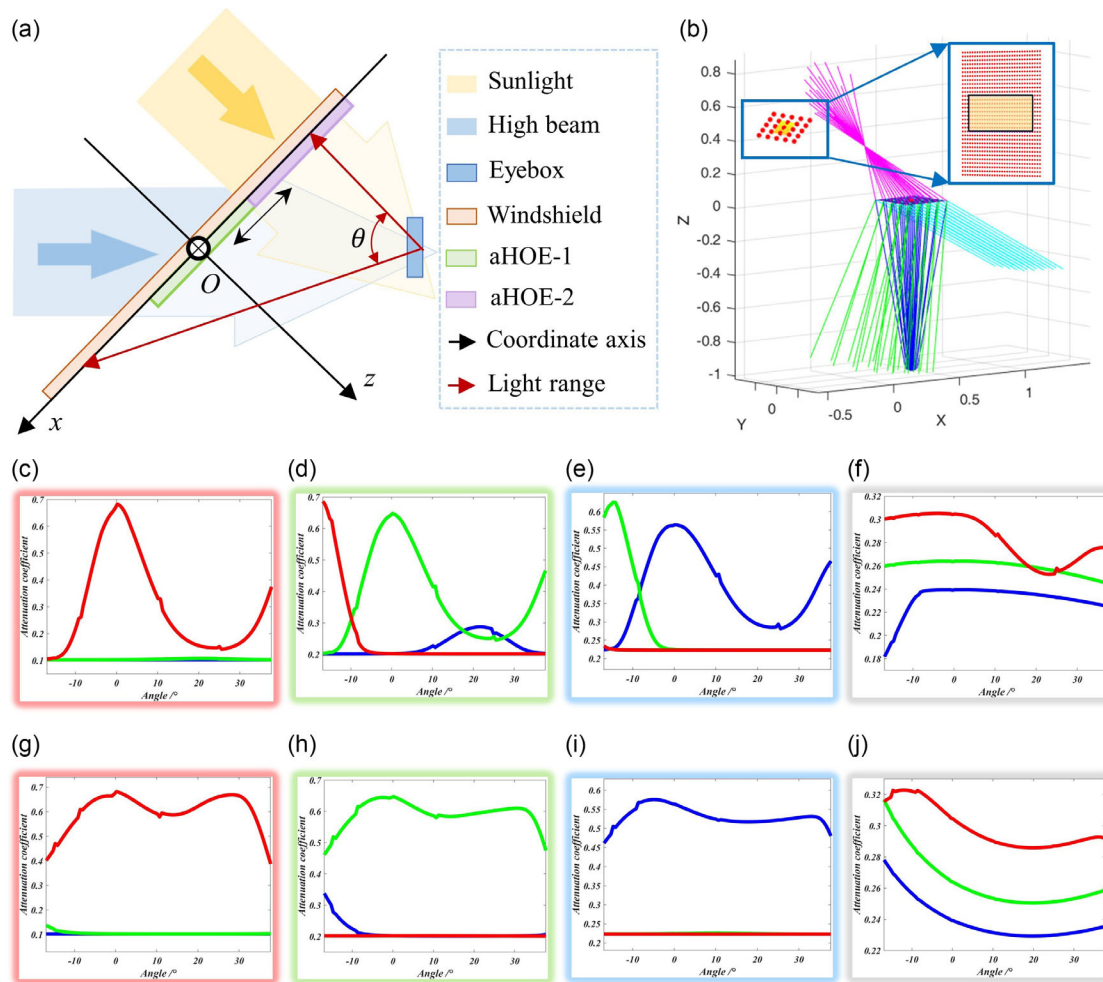


Figure 2. The design of aHOE is a) A simplified model of aHOE used for simulation, and the meanings of each part are shown in the legend on the right. HOE-1 and HOE-2 are two representative examples of aHOE at different positions, which play a role in detecting strong light at corresponding positions and angles. The design aims to determine the position of the recording point source and the wavelength of the recording light for each aHOE, so that it has a balanced attenuation effect on a wide spectrum of incident light across the entire angle range. b) A simulation model diagram of a certain position and wavelength, where the incident light landing point near the eye box is magnified in the upper right corner. c–f) A schematic diagram of the attenuation coefficient of external incident light in different bands as a function of incident angle when the aHOE at each position has the same recording setting. g–j) A schematic diagram of the attenuation coefficient of external incident light in different bands as a function of incident angle when each position's aHOE has different recording settings and all positions share the same set of recording point sources. The red, green, and blue curves in the figure represent the center wavelength of the aHOE recording light, while the color of the image border represents different bands of reconstructed light.

The enlarged image of the eye box plane is shown in the upper right corner, and the design process only considers the light rays with landing points within the eye box range.

To select a better recording method, two different schemes were used to simulate the variation of the attenuation coefficient of external incident light in different bands with the incident angle. In Method 1 (middle row of Figure 2), each position's aHOE has the same record setting, while in Method 2 (bottom row of Figure 2), each position's aHOE has different record settings, and all positions' aHOEs share the same set of recording point sources. The red, green, and blue curves in the figure represent the center wavelengths of aHOE recording light at 639, 532, and 457 nm, respectively. The colors of the image borders represent different wavelength ranges of incident light, with red,

green, blue, and gray representing $609 \approx 669$, $502 \approx 562$, $427 \approx 487$, and $400 \approx 700$ nm, respectively. The simulation results indicate that the attenuation effect is better when the wavelength range of the incident light is closer to the recording wavelength. Considering the incident light in the entire visible light range, when wavelength mismatch occurs, it may still approach the Bragg condition in other directions different from the recording angle and be diffracted with higher efficiency. Therefore, the attenuation coefficient is relatively gentle throughout the entire angle range. In addition, comparing the two schemes, Scheme 2 has a better attenuation effect, indicating that it is beneficial to design the aHOE separately for each position. However, it requires independent and controllable aHOE for each part, which requires higher system complexity. In the preliminary

system, if Scheme 1 is adopted for the sake of simplicity of the device, its attenuation effect is also acceptable.

The above is the design of the main functional aHOE in the system, and its effectiveness will be verified through experiments.

2.3. The Working Effect of AHOE

Due to the fact that daytime external light mainly comes from the sun and nighttime light has the highest proportion of LED, experiments are conducted on these two typical light sources.

2.3.1. LED Glare Source

To test the attenuation effect of aHOE on LED light, the test light path shown in **Figure 3a** is constructed. The lens in front of the LED flashlight is used to adjust the divergence of the light beam. To simplify the experiment without losing generality, parallel light is used for testing in this experiment. The aHOE is fixed on a bracket, and the rotating stage below it controls the angle of external light entering the aHOE. The light screen behind the aHOE is the plane where the driver's eye box is located,

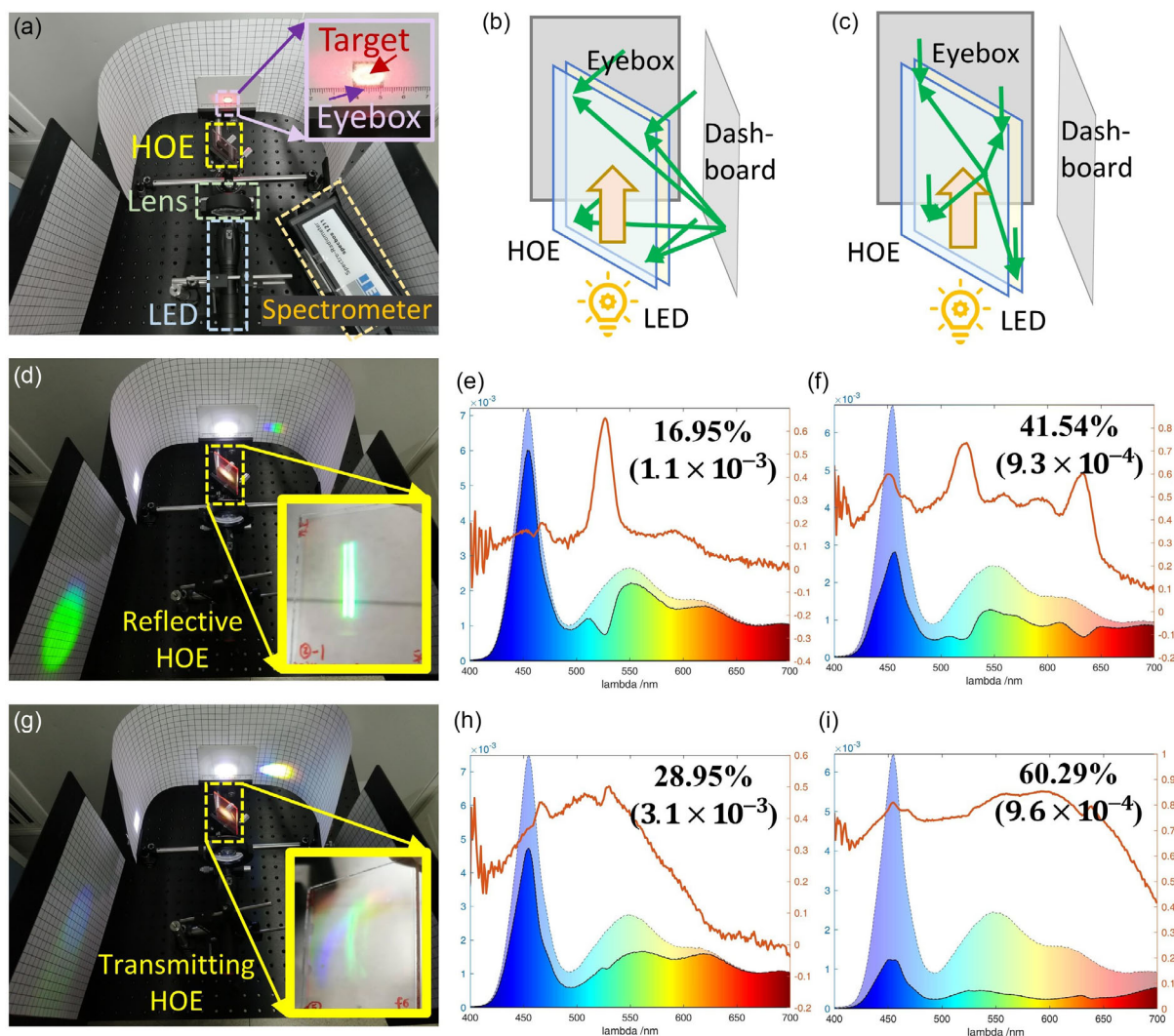


Figure 3. The functional testing experiment of aHOE when using LED flashlight as the light source. a) The optical path diagram of the testing system. b,c) Schematic diagram of the orientation when the reflective HOE and the transmissive HOE are located in the testing system, respectively. d) The light distribution of the reflective HOE illuminated by LED, and the enlarged image of the reflective HOE is shown in the lower right corner, where the green light is the diffraction image of the fluorescent lamp. e,f) The spectra of the light in the eye box before and after attenuation when using monochromatic aHOE and three-color multiplexed aHOE, respectively. The dark yellow curve represents the attenuation ratio at the corresponding wavelength, the number in the upper right corner represents the average of the total attenuation rate measured multiple times, and the number in parentheses represents the mean square error of the total attenuation rate. g) Represents the light distribution of a transmissive HOE illuminated by LED, and the enlarged image of the transmissive HOE is shown in the lower right corner, where the rainbow pattern represents its diffracted light under a fluorescent lamp. h,i) The spectra of the light in the eye box before and after attenuation when using monochromatic aHOE and three-color multiplexed aHOE, respectively. The dark yellow curve represents the attenuation ratio at the corresponding wavelength, the number in the upper right corner represents the average of the total attenuation rate measured multiple times, and the number in parentheses represents the mean square error of the total attenuation rate.

and the range of the eye box is marked with a red rectangle. A background plate with a grid around the light screen is used to roughly estimate the position of diffracted light. Place the spectrometer on the side and aim the target at the center of the eye box to measure the spectra before and after attenuation. Figure 3b and c show the correspondence between the testing system and the actual layout inside the car. The testing system simulates the scenario where the high beam is parallel to the windshield and enters the driver's eye box, with the dashboard located on the right side. The green lines represent the recording light, and the direction of the lines in the figure is only used to distinguish between reflective or transmissive HOEs. The experimental results indicate that reflective HOE is more suitable for antiglare function, but transmissive HOE has the ability to converge external light, making them feasible for expanding the light collection function of IHAWS.

Figure 3d shows the illumination distribution of a reflective HOE (5×5 cm) illuminated by incident light. Except for a portion of the light that is reflected and absorbed, most of the light that is not transmitted is diffracted to the lower left corner of the figure. As this reflective HOE uses light with a wavelength of 532 nm for recording, the wavelength of the diffracted light is mainly distributed around the recording wavelength. When aHOE rotates, the spectral components of the diffracted light will also change accordingly. On the right side of the screen is the stray light generated by the incident light passing through the aHOE, which is weak and does not enter the eye box range. Figure 3e shows the spectral attenuation of aHOE recorded at 532 nm, where the lighter colored spectrum is the light source spectrum before attenuation. To eliminate the influence of the substrate, a blank substrate is placed at the position of aHOE during the testing of this spectrum. The darker colored spectrum is the attenuated spectrum. To visually compare the attenuation effect at different wavelengths, the attenuation ratio curve is shown in dark yellow in the same graph. This curve indicates a significant attenuation peak near 532 nm, which is consistent with the theoretical analysis results. In order to analyze the repeatability of the test data, the attenuation rate curve is the average of 9 measurements, and the average total attenuation rate is 16.95%. The number in parentheses indicates that the mean square error of the total attenuation rate of multiple measurement data is 1.1×10^{-3} . To improve the attenuation ratio, single wavelength aHOE is replaced with a multiplexed aHOE recorded sequentially at 639, 532, and 457 nm.

Figure 3f shows the spectral attenuation of multiplexed aHOE, and significant attenuation peaks appear near all three recording wavelengths. Due to the nonuniform spectrum of LED, there is a high intensity in the blue light band, and the presence of blue gratings greatly attenuates the blue light component, which is not only beneficial for using limited multiplexed gratings to attenuate incident light to the maximum extent, but also reduces the harm of blue light to the human eye. The total attenuation rate of this aHOE over the entire spectral range is 41.54%.

To compare with the reflective HOE, the spectral attenuation plots of monochromatic transmission HOE and multiplexed transmission HOE are also measured in Figure 3h and i. There is no obvious attenuation peak in the transmissive HOE, mainly because the wavelength selection range of the

transmissive HOE is larger than that of the reflective one. Although the total attenuation rate of the transmissive HOE is better than that of the reflective HOE under the same conditions, it is not suitable as the main functional component for antiglare. The reason is that the diffracted light of the transmissive HOE will become stray light at other angles and enter the eye box range, which will also interfere with the driver. It is more suitable to be installed in the windshield area in front of the passenger seat for the collection of light energy. The test scenario of transmitting HOE shown in Figure 3g shows that most of the incident light is diffracted by HOE to the position of the dashboard. The energy of diffracted light is not only very concentrated, but its spectral unfolding can provide convenience for light collection. If photovoltaic panels are installed at the dashboard position, external light can be efficiently collected. Excluding reflected light and stray light, the collection ratio can reach nearly 60.29%. Moreover, as long as there is a light source, even during parking, the light collection device can work normally.

The above experimental results indicate that the designed reflective HOE can achieve over 40% attenuation of LED light in the entire visible light spectrum, which will greatly alleviate the interference of glare on drivers. In addition, the designed transmissive HOE can achieve about 60% light energy collection of incident light in the entire visible light band, which is attractive for light collection devices.

2.3.2. Solar Glare Source

To test the attenuation effect of aHOE on sunlight, a test optical path is constructed as shown in Figure 4a. Due to the fact that sunlight can provide a wider aperture of incident light compared to flashlights, the aHOE used is 30×30 cm, which is more suitable for practical applications. The reflective HOE is fixed on an angle adjustment bracket, and different solar elevation angles are simulated through the adjustment of the bracket. The spectrometer is installed on the rear side of the aHOE, with its target facing the center of the eye box. In order to prevent stray light from interfering with the test results, a light shield as shown in Figure 4b is made. A white light screen is used at the back to receive the light transmitted through the aHOE, and black material is attached to the inner side to absorb stray light. Figure 4c shows the appearance of the complete testing system with aHOE and light shield installed.

The middle row of Figure 4 measures the spectrum in the range of $350 \sim 1000$ nm, where the visible spectrum is represented by color and the near ultraviolet and near infrared spectra are indicated by gray. The lighter colored spectrum is the spectrum of light passing through a blank substrate, which can be approximated as the solar spectrum. Due to the weaker attenuation of long-wavelength light during atmospheric transmission, the near-infrared spectral component will be stronger, and the downward depression indicates the possible absorption of a certain component in the atmosphere. The darker colored spectrum is the spectrum attenuated by reflective HOE. Due to the fact that the aHOE is multiplexed from 639, 532, and 457 nm, the attenuated spectrum will have noticeable notches near the corresponding recording wavelengths. The upper left corner of each graph indicates the total attenuation ratio within the spectral

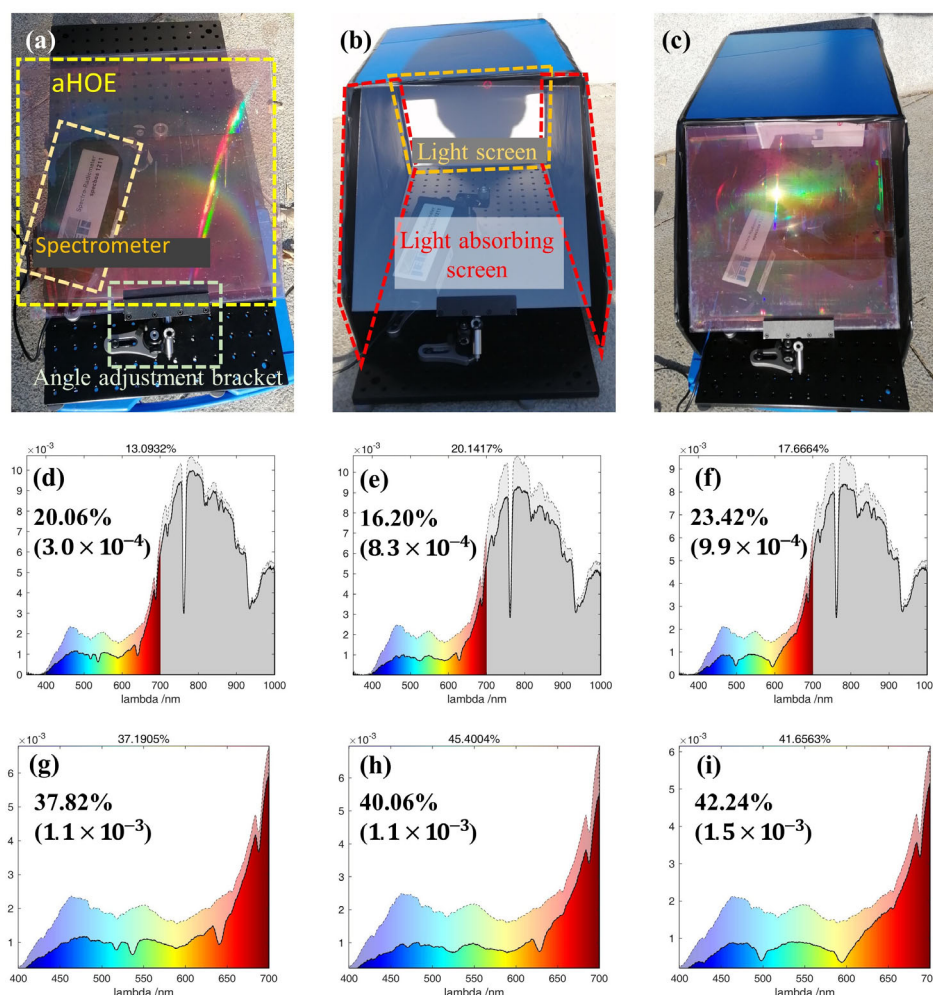


Figure 4. Functional testing experiment of aHOE when using the sun as the light source. a) The main structure of the testing system, aHOE is installed on the angle adjustment bracket, and the spectrometer is facing the center of the eye box. b) A light shield used to prevent interference from stray light, which only retains aHOE as a sunlight vent to prevent ambient light from entering. The eye box adopts a white light screen to receive transmitted light, and black material is attached to the side to absorb the stray light generated by reflection and scattering. c) The complete appearance of the testing system after installing the aHOE and adjusting it to a medium angle. d–f) The spectra of the light in the eye box before and after attenuation at different angles of sunlight incidence, with a spectral range of 350–1000 nm, covering the near ultraviolet to near infrared bands. The number in the upper left corner shows the average of the total attenuation ratio within the spectral range, and the number in parentheses represents the mean square error of the total attenuation rate. g–i) The spectra of the light in the eye box before and after attenuation at different angles of sunlight incidence, with a spectral range of 400–700 nm, only covering the visible light band. The number in the upper left corner shows the average of the total attenuation ratio within the spectral range, and the number in parentheses represents the mean square error of the total attenuation rate.

range. Although the attenuation results vary slightly at different incident angles, they are generally around 20%, and the mean square error is on the order of 10^{-4} . Due to the fact that the aHOE used is recorded using visible light, the attenuation in the near-infrared band may be relatively weak. However, in terms of antiglare function alone, the attenuation of near-infrared light that cannot be detected by the human eye is not a concern.

The antiglare function mainly examines the attenuation of light in the visible light band by aHOE. Therefore, in the lower row of Figure 4, the spectrum of the visible light band is extracted separately, with each column corresponding to the same incident light angle. As shown in the upper left corner of Figure 4g–i, the total attenuation rate fluctuates around 40% at different angles, with a mean square error of the order of 10^{-3} . This is similar to

the attenuation effect of aHOE on LEDs and can effectively suppress glare. In addition, near-infrared light has a stronger thermal effect compared to visible light. By adjusting the recording wavelength of aHOE to the near-infrared band, antiglare HOE can be converted into antihigh-temperature HOE, which is beneficial for slowing down the temperature rise inside the automobile.

2.4. Scheduling Scheme of Intelligent System

The above experiment demonstrated the antiglare effect of aHOE on several typical light incident directions. However, in practical applications, the glare source may come from multiple directions, which requires dynamic controllability of aHOE. The

previous text has proposed two design schemes for different positions of aHOE in the front windshield, which are applicable to different control mechanisms. The hHOE is usually fixed in the area directly facing the driver in the front windshield of a car, responsible for projecting driving information to the front of the vehicle to enhance driving safety. The aHOE and hHOE serve driving safety from the perspectives of eliminating obstacles and enhancing information, respectively. However, when the effective area of aHOE overlaps with hHOE, it introduces crosstalk to the displayed image, which actually weakens the original function of HUD. Therefore, in order to enable aHOE and hHOE to function separately without interfering with each other, it is necessary to provide an intelligent scheduling scheme for holographic antiglare window systems with HUD.

The flowchart of the proposed intelligent scheduling scheme is shown in Figure 5a, which aims to provide the positions of aHOE and hHOE without overlapping for any angle of the glare source. Firstly, determine the orientation of the glare source, then calculate the working area of aHOE, and then determine whether the working areas of aHOE and hHOE overlap. If they do not overlap, aHOE and hHOE will work separately and unleash their full potential. Otherwise, the position of aHOE remains unchanged, while hHOE only enables projection of non-overlapping areas. Below is a detailed analysis of the glare source orientation detection module and the AR zoned display module.

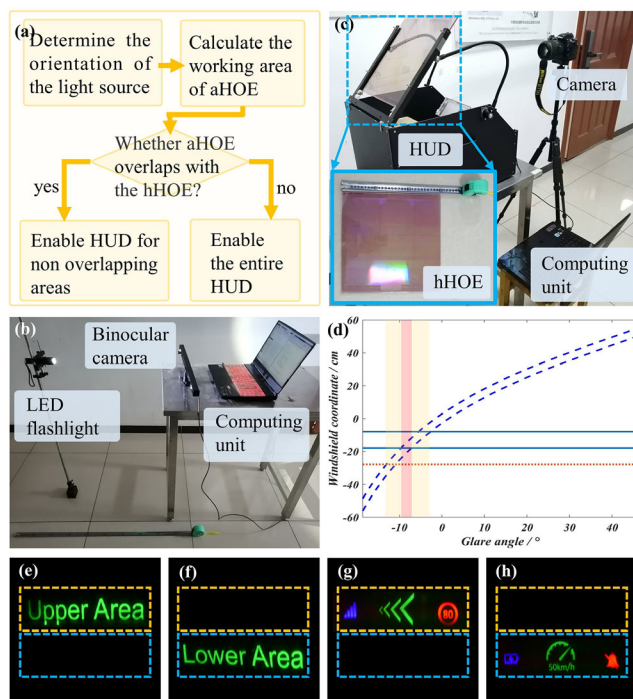


Figure 5. Scheduling scheme for intelligent systems. a) Flowchart of intelligent scheduling algorithm. b) Experimental setup for glare source depth detection based on binocular cameras. c) Experimental setup for capturing the AR display effect of HUD. d) Simulation results of the displacement of aHOE and hHOE positions when the incident angle of the glare source changes. e, f) Illustrate the AR display effects in the upper and lower areas of the hHOE, respectively. g, h) Demonstrate the full-color AR display effects in the upper and lower areas of the hHOE, respectively.

Figure 5b shows the device diagram of the glare source orientation detection module. Firstly, movable LED flashlights are used to simulate glare sources at different locations. Secondly, binocular cameras are used to capture photos containing glare sources and transmit them to the computing unit for processing. Finally, the image processing method is used to obtain the position coordinates of the glare source relative to the binocular camera (more details can be found in Section S1, Supporting Information).

To showcase the zoned display capabilities of hHOE, the system shown in Figure 5c is constructed. The hHOE is mounted on a tilted bracket to simulate the tilt angle of the windshield, and its front view is enlarged and displayed in the lower left corner. The camera is installed at the position of the driver's eye box to capture the scene observed by the driver. The picture generation unit (PGU) in the black box provides the image source for hHOE. The computing unit adjusts the image source of PGU based on the results of the glare source orientation detection module, ensuring that only non-overlapping areas of hHOE receive the projected image. Figure 5d is a simulation diagram of the displacement of aHOE as a function of glare incidence angle, providing a basis for the zoned display of HUD, which needs to be calculated based on HUD parameters such as field of view and eye box. To illustrate the principle without loss of generality, this paper selects a plane parallel to the xoz plane and passing through the midpoint of the eye box for simulation. The horizontal axis represents the angle range of the incident glare, where zero degree represents the horizontal incident light, and the positive and negative angles represent the light incident from top to bottom and from bottom to top, respectively. The vertical axis represents the position coordinates on the windshield, zero represents the position of the horizontal centerline of the windshield, and positive and negative represent the positions above and below the centerline, respectively. The two blue dashed lines represent the positions of the upper and lower edges of aHOE. Since the simulation assumes that the incident light is plane light, which requires the largest aHOE area, the area between these two blue dashed lines is the maximum range that aHOE can occupy. The range between the two blue solid lines represents the range of the upper half of the hHOE, while the range between the lower blue solid line and the yellow dotted line represents the range of the lower half of the hHOE. It can be seen that the areas aHOE and hHOE outside the color blocks in the figure do not overlap and can work separately. In the area of two yellow blocks, aHOE will overlap with one of the upper or lower halves of hHOE, and only the non-overlapping area of hHOE will be used for display. In the area of the red color block, aHOE will overlap both the upper and lower areas of hHOE. At this point, different strategies can be selected according to the user's needs. On the one hand, simple navigation information can be displayed in the upper quarter and lower quarter of the area. On the other hand, since the angle of the glare source will change with time and driving direction, and the area occupied by the red color block is only 4% of the entire angle range, the HUD display can be temporarily turned off in this situation and restarted after the glare source angle changes. Figure 5e and f show the AR display effects of the upper and lower areas of hHOE, respectively. The green color represents display information, while the yellow and blue dashed boxes

are used to clearly indicate the range of the two areas. Figure 5g and h, respectively, demonstrate the color display capabilities of the upper and lower areas of hHOE.

The above scheduling scheme provides a method to avoid overlap between aHOE and hHOE, which can ensure the normal operation of HUD while achieving antiglare. In addition, for high-speed vehicles, the robustness of the antiglare system is important. In terms of tracking stability, although the position difference of the light source relative to the binocular camera and the human eye can usually be ignored, in order to further improve stability, the deployment of multiple cameras can be considered in future work. In terms of tracking speed, the i7-10870H CPU @ 2.20 GHz processor was used to run the program in the experiment, and the time required for a single image processing was 1.6 s, which is sufficient in the prototype. However, in order to improve product maturity, on the one hand, dedicated chips can be designed to enhance computing speed. On the other hand, by drawing on research such as programmable metasurfaces,^[31] new materials or devices can be used to achieve a refreshing aHOE with response speeds on the order of hundreds of microseconds.

3. Conclusion

This paper proposes an antiglare holographic automotive windshield system. The aHOE in the system can achieve a light energy attenuation of about 40% for LED and sunlight, which is beneficial for avoiding glare interference to the driver and improving driving safety. In addition, intelligent scheduling algorithms are used to separate the working areas of aHOE and hHOE, ensuring that HUD is not affected by crosstalk. The innovative contribution of this paper is primarily manifested in the application of HOE to design an antiglare system for HUD-integrated automobiles. Furthermore, the designed aHOE can balance the antiglare effect in a wide angle range and a wide spectral range. Given the design flexibility and scalability of IHAWs, there will be many interesting studies to be conducted in the future. The first is to combine HOE with refreshable devices or mechanical components to achieve fully automated adjustment of antiglare systems. The second is to combine the transmissive aHOE collector with photovoltaic panels to construct a solar power generation device that provides electrical energy for the vehicle's electronic system. The third is to adjust the recording wavelength of the aHOE, extending the antiglare HOE to antiultraviolet HOE operating in the near ultraviolet band or antihigh-temperature HOE operating in the near infrared band. It is worth mentioning that the aHOE in IHAWs can be applied not only in automobiles, but also in glass exterior walls of buildings, providing harmful light protection and energy collection for residential buildings. It is believed that the IHAWs proposed in this paper can effectively reduce the danger caused by glare and provide new ideas for the safety design of automobiles.

4. Experimental Section

Fabrication of HOE: A holographic exposure system was built to produce hHOE and aHOE. The laser combined by red, green, and blue wavelengths

was divided into two beams by a polarization beam splitter, which were filtered by a pinhole filter, respectively. Then, as required, the beam was collimated into a plane wave by a collimating lens, or the wavefront of the recording light was shaped by a wavefront modulation element such as a lens (group). The holographic exposure material (photopolymer) was fixed and ensured that the light spots of the two recording beams coincided on the holographic exposure material. The power of the red, green, and blue lasers were adjusted separately and it was ensured that each wavelength can achieve sufficient exposure (15 mJ/cm² for red laser, 20 mJ/cm² for green laser, and 25 mJ/cm² for blue laser). UV curing was performed on the exposed HOE under a UV lamp. The above were the similarities between aHOE and hHOE in fabrication, but their differences lay in the shape of the recording wavefront and the angle of incidence, thus requiring different optical paths.

Directional detection of glare sources: The image processing method based on binocular cameras was adopted to obtain the position coordinates of glare sources. A 3D right-handed coordinate system was established with the midpoint of the left and right cameras in the binocular camera as the origin, pointing from the origin to the right camera as the positive y-axis direction, and the opposite direction of the camera's shooting direction as the positive z-axis direction. The computing unit was used to control the binocular camera to capture images containing glare sources. The checkerboard was used to calibrate the camera's intrinsic and extrinsic parameters, and the captured original image was subjected to epipolar correction based on the calibration results. Image processing was used to obtain glare source masks (more details can be found in Section S1, Supporting Information) from the original images of the left and right cameras, respectively. The masks contained the position information of the glare source. The disparity d of the glare source in the left and right images was calculated by the centroid of the masks. The depth information of the glare source was calculated according to the formula $z = f \cdot B/d$, where f and B are the focal length and baseline length of the camera, respectively. The coordinates of the glare source in the 3D coordinate system were calculated by combining the position and depth information of the glare source.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

antiglare, head-up displays, holographic optical elements, intelligent systems

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